Constraint-Aware Hierarchical Risk Parity for Robust Portfolio Allocation

Pramukh Rajendra Jain
School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India
pramukhjain.9966@gmail.com

November 2026

Abstract

Hierarchical Risk Parity (HRP) is a popular portfolio construction method that allocates capital through a correlation tree instead of inverting a covariance matrix, which makes it robust to estimation error. However, HRP cannot respect the constraints that real mandates impose: per-asset limits, sector or asset-class caps and floors, and turnover budgets. We propose Constraint-Aware HRP (CA-HRP), which keeps HRP's hierarchical risk allocation as a target and maps it to the feasible portfolio with minimum ex-ante tracking error to it. We prove that this projection is unique, leaves compliant HRP portfolios unchanged, and redistributes excess weight as a covariance-based hedge. We also prove that a solver-free capacity-aware bisection always meets per-asset bounds exactly, and we study a regime-robust projection with a worst-case guarantee over a convex set of covariance matrices. The design is fixed on a validation period and then tested on a held-out period, on three universes added only after the method was frozen (Indian ETFs, US sectors and global country funds), at three rebalancing frequencies, and in a Monte Carlo study with a known covariance — six universes and more than 1,300 monthly rebalances from 2001 to 2026 in total. CA-HRP satisfies every constraint at every rebalance, whereas HRP breaches them at 48–100% of rebalances and the common clip-and-renormalise fix at up to 46%. Where constraints bind hardest, CA-HRP raises the Sharpe ratio from 0.76 to 0.91 relative to Euclidean projection, with 25% lower volatility; it has the lowest or equal-lowest volatility of the HRP fixes in all six universes and the highest true Sharpe ratio of all methods in the simulation. It does not, however, beat constrained minimum variance on realised returns, and the regime-robust variant did not help in the 2022 stock–bond correlation shift. Most gains come from group constraints, and the risk-aware projection metric, not the estimator or clustering rule, drives them. We further prove that the projection never amplifies estimation noise, whereas clipping amplified it up to tenfold in our data, and we give an exact, shadow-price decomposition of the "price of compliance" that identifies group limits as responsible for 67–82% of it wherever it is material. In 160 random mandates, including a scrambled-group placebo, CA-HRP stayed compliant everywhere and had lower or equal volatility than Euclidean projection in 85% of cases, but its return advantage was confined to correlation-aligned group constraints. A forward-looking variant that rescales volatilities with option-implied volatility indices (VIX, VXN, GVZ, OVX, MOVE, India VIX) softened some crashes but did not change risk-adjusted returns and doubled turnover. An eleven-check verification suite, including an exact match with an independent HRP implementation and a look-ahead test, accompanies the code.

Keywords: hierarchical risk parity, portfolio optimisation, constrained allocation, tracking error, robust estimation, convex optimisation.

1 Introduction

Mean–variance optimisation [12] is the textbook answer to portfolio construction, yet in practice its outputs are notoriously unstable: small errors in estimated means and covariances are amplified by the inversion of an ill-conditioned covariance matrix into extreme, concentrated and rapidly

changing weights [3, 14]. Out of sample, optimised portfolios frequently fail to beat naive equal weighting [4]. Two families of remedies have emerged. The first regularises the inputs, for example by shrinking the covariance matrix [8] or by imposing weight constraints, which Jagannathan and Ma [7] show act as a form of shrinkage. The second abandons optimisation altogether in favour of heuristic risk allocation, such as equal risk contribution [11].

Hierarchical Risk Parity (HRP), introduced by López de Prado [10], belongs to the second family and has become one of the most popular allocation methods among practitioners and in open-source libraries. HRP clusters assets by correlation, orders them so that similar assets are adjacent, and allocates capital top-down by recursively splitting each cluster in inverse proportion to its risk. Because it never inverts the covariance matrix, HRP is robust to estimation noise and can handle singular covariance matrices.

HRP has, however, a practical blind spot: it cannot respect constraints. Real mandates impose per-asset limits (UCITS concentration rules, single-stock caps), sector or asset-class caps and floors (investment policy statements), and turnover budgets (to control trading costs). Recursive bisection has no mechanism for any of these, and in our multi-asset universe unconstrained HRP places on average 86% of capital in a single short-term bond fund, breaching a 15% single-fund cap and a 40% asset-class cap at *every* rebalance. The usual fixes are ad hoc: weights are clipped and renormalised, which can cycle without ever reaching the feasible set, or projected onto it with a Euclidean distance that ignores how assets co-move.

This paper proposes **Constraint-Aware Hierarchical Risk Parity (CA-HRP)**, a principled way to keep HRP's risk allocation while satisfying any set of linear constraints exactly. Our contributions are:

1. **Capacity-aware recursive bisection.** We define a capacity interval for every node of the HRP tree and clip each split into it. We prove that the interval is never empty and that per-asset bounds are met exactly, with no solver (Proposition 1). This yields a solver-free variant, CA-HRP-B.

2. **Tracking-error projection.** We map the HRP target to the feasible portfolio with minimum ex-ante tracking error to it. We show the projection is unique, leaves compliant HRP portfolios unchanged, and redistributes excess weight as a covariance-based hedge rather than uniformly (Proposition 2).

3. **A transparent evaluation protocol.** The method design was selected on a validation period only and then frozen; we report a separate held-out test period, all constraint sets were fixed before any backtest was run, and all code and data are public.

4. **Evidence on six universes and in simulation.** On 20 multi-asset ETFs (2008–2026), 43 NIFTY 50 stocks (2011–2026) and 47 US large caps (2006–2026), and on three external-validation universes added after the method was frozen (Indian ETFs, US sector funds and global country funds), CA-HRP satisfies every constraint at every rebalance and at every rebalancing frequency. Where constraints bind, it is the strongest member of the HRP family; where they rarely bind, it matches HRP. In a Monte Carlo study with a known covariance it has the highest true Sharpe ratio of all methods.

5. **Explaining the method.** We prove that the projection is non-expansive, so it can never amplify estimation noise (Proposition 3), and show empirically that clipping does amplify it. We derive an exact identity that splits the cost of compliance over constraints through their shadow prices (Proposition 4) and use Shapley values to attribute it to the rules of a mandate.

6. **Forward-looking risk.** We feed option-implied volatility into CA-HRP through implied-to-realised ratios of six volatility indices, prove that only cross-asset differences in these ratios can matter (Proposition 6), and report its mixed effect.

7. **Stress tests and verification.** We test 160 random mandates with a scrambled-group placebo, apply the Model Confidence Set and the deflated Sharpe ratio, check risk-forecast calibration, and publish an eleven-check verification suite.

8. **A regime-robust variant and two negative results.** We propose a worst-case tracking-error projection over covariance scenarios with a guarantee over their convex hull (Proposition 5), and report honestly that it does not improve on CA-HRP, and that embedding constraints inside the bisection performs worse than projecting afterwards.

The rest of the paper is organised as follows. Section 2 reviews related work, Section 3 states the problem, Section 4 presents the method and its properties, Section 5 describes the data and experimental protocol, Section 6 reports the main results, Section 7 the extended evaluation, Section 8 explanation, stress tests and verification, and Section 10 concludes.

2 Related Work

Estimation error and robust allocation. The sensitivity of mean–variance portfolios to input error is well documented [3, 14]. DeMiguel et al. [4] find that across many data sets none of fourteen optimised models consistently beats $1/N$ out of sample. Shrinkage estimators [8] reduce the condition number of the covariance matrix, and Jagannathan and Ma [7] show that no-short-sale constraints behave like covariance shrinkage. Risk-based methods allocate by risk rather than expected return; the equal-risk-contribution (ERC) portfolio [11] equalises marginal risk contributions and can be computed by convex optimisation [21].

Hierarchical allocation. López de Prado [10] introduced HRP, combining graph-theoretic clustering with recursive inverse-variance bisection. Raffinot [17] studied hierarchical clustering-based allocation more broadly, and Raffinot [18] proposed the Hierarchical Equal Risk Contribution (HERC) portfolio, which splits along the dendrogram rather than the ordered list. Pfitzinger and Katzke [15] recognised that HRP lacks constraint handling and proposed a cluster-based capital allocation that accommodates weight bounds within the bisection.

Positioning of this work. Capacity-aware bisection is closest in spirit to Pfitzinger and Katzke [15]; our contribution is a capacity definition that also incorporates group caps, together with a short proof of non-emptiness and exact box feasibility based on the additivity and sub-additivity of the capacity bounds. More importantly, we show empirically that enforcing constraints *inside* the bisection has a hidden cost: excess weight cascades down one branch and can concentrate in volatile assets. Our main method therefore separates the two concerns: HRP supplies the risk allocation, and a tracking-error projection [19] enforces arbitrary linear constraints (group floors, caps and turnover) with minimum risk distortion. To our knowledge this is the first study to compare constraint-handling strategies for HRP under a common, pre-registered protocol with a held-out test period.

3 Problem Setting

We consider N risky assets with daily simple returns $r_t \in \mathbb{R}^N$. At each rebalancing date the manager observes the last L return vectors, estimates a covariance matrix $\hat{\Sigma}$, and chooses weights $w \in \mathbb{R}^N$. The mandate imposes a polyhedral feasible set

$$\mathcal{W}(w^-) = \Big\{ w : \ \mathbf{1}^\top w = 1, \ \ \ell \le w \le u, \ \ \underline{c}_g \le \sum_{i \in g} w_i \le \bar{c}_g \ \ \forall g \in \mathcal{G}, \ \ \|w - w^-\|_1 \le \tau \Big\}, \qquad (1)$$

where ℓ, u are per-asset bounds, $\mathcal{G}$ partitions the assets into groups (sectors or asset classes) with caps $\bar{c}_g$ and floors $\underline{c}_g$, w^- is the drifted pre-trade portfolio and τ a turnover budget. All constraints

are linear, so $\mathcal{W}$ is a convex polytope. We assume it is non-empty. The turnover budget can in principle conflict with the other constraints if price drift pushes the pre-trade portfolio far outside the box or group limits; our implementation then relaxes the turnover cap for that date only. In all experiments reported here this fallback was never triggered: every CA-HRP portfolio satisfied all constraints, turnover included.

4 Method

4.1 Hierarchical Risk Parity

HRP [10] proceeds in three steps. (i) *Tree clustering*: the correlation matrix ρ implied by $\hat{\Sigma}$ is mapped to the distance $d_{ij} = \sqrt{(1-\rho_{ij})/2}$ and agglomerated with single linkage. (ii) *Quasi-diagonalisation*: assets are ordered by the dendrogram leaves so that similar assets are adjacent. (iii) *Recursive bisection*: starting from the full ordered list with weight $W = 1$, every cluster C is split into a left and right half C_L, C_R; with $V_S = \tilde{w}_S^\top \hat{\Sigma}_{SS} \tilde{w}_S$ the variance of the inverse-variance portfolio $\tilde{w}_S \propto \mathrm{diag}(\hat{\Sigma}_{SS})^{-1}$ of a cluster S, the split fraction is

$$\alpha_C = 1 - \frac{V_{C_L}}{V_{C_L} + V_{C_R}}, \qquad W_{C_L} = \alpha_C W_C, \quad W_{C_R} = (1-\alpha_C) W_C. \tag{2}$$

The recursion ends at single assets. No matrix is inverted and every split is a two-asset risk-parity decision, which is the source of HRP's stability. Nothing in (2), however, knows about $\mathcal{W}$.

4.2 Capacity-aware recursive bisection (CA-HRP-B)

Our first contribution makes the bisection itself respect per-asset bounds and, partially, group caps, without any optimisation solver. For a set of assets S define its *capacity interval*

$$\mathrm{lo}(S) = \sum_{i \in S} \ell_i, \qquad \mathrm{hi}(S) = \sum_{g \in \mathcal{G}} \min\Big(\sum_{i \in S \cap g} u_i,\ \bar{c}_g \Big). \tag{3}$$

$\mathrm{lo}(S)$ is the least weight that S must hold and $\mathrm{hi}(S)$ an upper bound on what it can hold. When a cluster C carrying weight W_C is split, the HRP fraction (2) is clipped to

$$\alpha_C \in \Big[\tfrac{\max(\mathrm{lo}(C_L),\, W_C - \mathrm{hi}(C_R))}{W_C}, \quad \tfrac{\min(\mathrm{hi}(C_L),\, W_C - \mathrm{lo}(C_R))}{W_C} \Big]. \tag{4}$$

Proposition 1 (Exact box feasibility)**.** *If* $\mathrm{lo}(\{1,\dots,N\}) \le 1 \le \mathrm{hi}(\{1,\dots,N\})$*, the interval* (4) *is non-empty at every split, and the resulting weights satisfy* $\mathbf{1}^\top w = 1$ *and* $\ell_i \le w_i \le \min(u_i, \bar{c}_{g(i)})$ *for all* i*.*

Proof. We show by induction that every cluster satisfies the invariant $\mathrm{lo}(C) \le W_C \le \mathrm{hi}(C)$. It holds at the root by assumption. Note that lo is additive, $\mathrm{lo}(C) = \mathrm{lo}(C_L) + \mathrm{lo}(C_R)$, and hi is sub-additive because $\min(a+b, c) \le \min(a,c) + \min(b,c)$ for $a, b, c \ge 0$, so $\mathrm{hi}(C) \le \mathrm{hi}(C_L) + \mathrm{hi}(C_R)$. The interval (4) (multiplied by W_C) is non-empty iff the four inequalities $\mathrm{lo}(C_L) \le \mathrm{hi}(C_L)$, $\mathrm{lo}(C_L) \le W_C - \mathrm{lo}(C_R)$, $W_C - \mathrm{hi}(C_R) \le \mathrm{hi}(C_L)$ and $W_C - \mathrm{hi}(C_R) \le W_C - \mathrm{lo}(C_R)$ hold. The first and last follow from $\ell \le u$; the second from $W_C \ge \mathrm{lo}(C)$ and additivity; the third from $W_C \le \mathrm{hi}(C)$ and sub-additivity. Any α_C in the interval gives children with $\mathrm{lo}(C_L) \le W_{C_L} \le \mathrm{hi}(C_L)$ and likewise for C_R, which preserves the invariant. At a leaf $\{i\}$, $\mathrm{lo} = \ell_i$ and $\mathrm{hi} = \min(u_i, \bar{c}_{g(i)})$, which proves the claim. Weights sum to one because each split conserves W_C. □

CA-HRP-B costs $O(N^2)$ beyond HRP and needs no solver. It guarantees box bounds exactly, but group constraints are only respected when a group lies inside one sub-tree, and group floors and turnover are not handled. Those remaining constraints are enforced by a light projection (next subsection, with the Euclidean metric replaced by a tree-weighted one; details in the code). As our experiments show, the price of handling constraints *inside* the bisection is that excess weight cascades down a single branch of the tree and can end up concentrated in volatile assets; this motivates the main method.

Algorithm 1 Constraint-Aware HRP (one rebalancing date)

Require: returns window $R \in \mathbb{R}^{L\times N}$, feasible set $\mathcal{W}$, drifted weights w^-

1: $\hat{\Sigma}_{\mathrm{LW}} \leftarrow$ Ledoit–Wolf shrinkage covariance of R
2: $\rho \leftarrow$ correlation of $\hat{\Sigma}_{\mathrm{LW}}$; $d_{ij} \leftarrow \sqrt{(1-\rho_{ij})/2}$
3: $\pi \leftarrow$ leaf order of the single-linkage dendrogram of d
4: $w^{\mathrm{HRP}} \leftarrow$ recursive bisection (2) on π ▷ optionally clipped by (4)
5: $w^{\mathrm{CA}} \leftarrow$ solve the convex QP (5) over $\mathcal{W}(w^-)$
6: **return** w^{CA}

4.3 Tracking-error projection (CA-HRP)

The key observation is that HRP produces an excellent *target* but has no notion of distance between portfolios. Practitioners close the gap by clipping and renormalising, which is not guaranteed to reach $\mathcal{W}$, or by a Euclidean projection $\min_{w\in\mathcal{W}}\|w - w^{\mathrm{HRP}}\|_2^2$, which treats a unit of weight moved from a short-term bond fund to an oil fund the same as one moved to another bond fund. We instead measure distance in *risk*: the ex-ante tracking-error variance of the constrained portfolio relative to HRP [19],

$$w^{\mathrm{CA}} = \arg\min_{w\in\mathcal{W}(w^-)} (w - w^{\mathrm{HRP}})^\top \bar{\Sigma}\,(w - w^{\mathrm{HRP}}), \qquad \bar{\Sigma} = \frac{N\hat{\Sigma}_{\mathrm{LW}}}{\operatorname{tr}\hat{\Sigma}_{\mathrm{LW}}} + \epsilon I, \tag{5}$$

where $\hat{\Sigma}_{\mathrm{LW}}$ is the Ledoit–Wolf shrinkage estimator [8], w^{HRP} is HRP computed on $\hat{\Sigma}_{\mathrm{LW}}$, and the small ridge $\epsilon = 10^{-4}$ guarantees strict convexity.

Proposition 2 (Properties of the projection)**.** *(a) If $\mathcal{W}(w^-) \neq \emptyset$, problem* (5) *has a unique solution. (b) If $w^{\mathrm{HRP}} \in \mathcal{W}(w^-)$ then $w^{\mathrm{CA}} = w^{\mathrm{HRP}}$: CA-HRP never alters HRP when HRP already complies. (c) Among all feasible portfolios, w^{CA} minimises the ex-ante variance of the return difference to HRP, so it is the feasible portfolio whose returns are most correlated with, and least volatile around, the HRP portfolio. (d) Let $Aw \le b$ collect the active inequality constraints and $\mathbf{1}^\top w = 1$ the budget. The KKT conditions give $w^{\mathrm{CA}} = w^{\mathrm{HRP}} - \frac{1}{2}\bar{\Sigma}^{-1}(A^\top\lambda + \nu\mathbf{1})$ with $\lambda \ge 0$: the correction is a* hedge. *Weight removed from a capped asset is redistributed in proportion to the rows of $\bar{\Sigma}^{-1}$, i.e. towards the assets that best replicate the capped asset's risk.*

Proof. (a) $\bar{\Sigma} \succ 0$, so the objective is strictly convex and $\mathcal{W}$ is a non-empty closed convex set; a unique minimiser exists. (b) The objective is non-negative and zero at w^{HRP}. (c) $(w - w^{\mathrm{HRP}})^\top\hat{\Sigma}(w - w^{\mathrm{HRP}})$ is the variance of $(w - w^{\mathrm{HRP}})^\top r$ under the estimated covariance; the trace normalisation and ridge only rescale and regularise it. (d) Stationarity of the Lagrangian $(w - w^{\mathrm{HRP}})^\top\bar{\Sigma}(w - w^{\mathrm{HRP}}) + \lambda^\top(Aw - b) + \nu(\mathbf{1}^\top w - 1)$, with the ℓ_1 turnover constraint written in its standard linear epigraph form. □

Part (d) explains the behaviour we observe in the multi-asset universe: when HRP wants far more than the 40% asset-class cap in bonds, the Euclidean projection spreads the excess almost uniformly (about 5% to each non-bond fund), while the tracking-error projection re-shapes the portfolio as a hedge: inside the bond budget it moves from short-term to intermediate and long Treasuries (IEF, TLT), whose negative correlation with equities offsets the risk of the assets that must now be held, and outside the bond budget it favours gold and broad commodities (Section 6). The constrained portfolio keeps the defensive character of HRP, with markedly lower volatility than the Euclidean alternative.

Complexity. Steps 1–4 cost $O(LN^2 + N^2\log N)$. Step 5 is a QP with N variables (plus N auxiliary variables for the turnover epigraph) and $N + |\mathcal{G}|$ linear constraints; with an interior-point solver [5] it takes a few milliseconds for $N \le 50$. A full 20-year monthly backtest of CA-HRP runs in under six seconds on a laptop; Section 7 reports timings up to $N = 400$.

4.4 Two further properties: noise damping and the price of compliance

Proposition 3 (Noise damping)**.** *Fix the metric $\bar{\Sigma} \succ 0$ and the feasible set $\mathcal{W}$, and let $P(t) = \arg\min_{w\in\mathcal{W}} \|w - t\|_{\bar{\Sigma}}$ with $\|x\|_{\bar{\Sigma}}^2 = x^\top \bar{\Sigma} x$. Then for any two targets t_1, t_2, $\|P(t_1) - P(t_2)\|_{\bar{\Sigma}} \le \|t_1 - t_2\|_{\bar{\Sigma}}$.*

Proof. $\langle x, y\rangle_{\bar{\Sigma}} = x^\top \bar{\Sigma} y$ is an inner product, so P is the metric projection onto a closed convex set in a Hilbert space, which is firmly non-expansive [2]: $\|Pt_1 - Pt_2\|^2 \le \langle Pt_1 - Pt_2,\, t_1 - t_2\rangle \le \|Pt_1 - Pt_2\|\,\|t_1 - t_2\|$ by the variational inequality characterising projections and the Cauchy–Schwarz inequality. □

Proposition 3 means that whatever estimation noise the HRP target carries, the projection can only shrink it, measured in the same risk metric. The clip-and-renormalise heuristic has no such property; Section 8 shows that in practice it often *amplifies* noise.

Proposition 4 (Exact price-of-compliance identity)**.** *Let $f^\star = (w^{\mathrm{CA}} - t)^\top \bar{\Sigma}(w^{\mathrm{CA}} - t)$ be the optimal value of* (5) *with linear constraints $a_k^\top w \le b_k$ and multipliers $\lambda_k \ge 0$. Then*

$$f^\star = \tfrac{1}{2} \sum_k \lambda_k \,(a_k^\top t - b_k), \tag{6}$$

so the cost of compliance splits exactly over constraints, each term being the shadow price of a constraint times the amount by which the HRP target t violates it.

Proof. Stationarity gives $2\bar{\Sigma} d = -(\sum_k \lambda_k a_k + \nu \mathbf{1})$ with $d = w^{\mathrm{CA}} - t$. Multiply by $d^\top/2$: $f^\star = -\frac{1}{2}\sum_k \lambda_k a_k^\top d$, because $\mathbf{1}^\top d = 0$. Complementary slackness gives

$$\lambda_k a_k^\top w^{\mathrm{CA}} = \lambda_k b_k, \qquad \text{hence} \qquad -\lambda_k a_k^\top d = \lambda_k (a_k^\top t - b_k).$$

□

Individual terms of (6) can be negative: a cap that the target already satisfies but that binds at the optimum contributes negatively. To report the cost by *family* of constraints (asset caps; group caps and floors; turnover) we therefore use the Shapley value [20] of the cooperative game $v(S) = f_S^\star$, the optimal value when only the families in S are imposed. It is order-independent and adds up exactly to $f^\star$, and it costs $2^3 = 8$ small solves per date. The ℓ_1 turnover cap is not linear in w, but in its standard linear epigraph form the identity still holds; we compute the turnover term as $f^\star$ minus the linear terms and use (6) as an exactness check on every date (the remainder must vanish whenever the turnover cap is slack).

4.5 Regime-robust projection (CA-HRP-R)

The tracking-error metric in (5) trusts a single covariance estimate. When correlations shift, the hedge chosen by the projection can fail. We therefore also study a *regime-robust* projection that tracks HRP under the worst of K covariance scenarios $\{\Sigma_k\}_{k=1}^K$:

$$w^{\mathrm{R}} = \arg \min_{w\in\mathcal{W}(w^-)} \; \max_{k=1,\dots,K} \; (w - w^{\mathrm{HRP}})^\top \bar{\Sigma}_k (w - w^{\mathrm{HRP}}), \tag{7}$$

where every $\bar{\Sigma}_k$ uses the same scale factor as the baseline estimate, so that relative risk levels across regimes are preserved. We consider three scenarios: the 1-year Ledoit–Wolf estimate, a 3-month (63-day) Ledoit–Wolf estimate capturing the current regime, and a "hedges-fail" stress covariance $D|\rho|D$ in which every correlation is replaced by its absolute value.

Proposition 5 (Robust projection)**.** *Problem* (7) *is convex, has a unique solution, and returns w^{HRP} whenever it is feasible. Its optimal value $t^\star$ bounds the ex-ante tracking-error variance under* every *covariance matrix in the convex hull* $\mathrm{conv}\{\bar{\Sigma}_1, \dots, \bar{\Sigma}_K\}$*:* $(w^{\mathrm{R}} - w^{\mathrm{HRP}})^\top \Sigma (w^{\mathrm{R}} - w^{\mathrm{HRP}}) \le t^\star$.

Proof. Each term is a strictly convex quadratic, and the pointwise maximum of strictly convex functions is strictly convex; with a non-empty convex feasible set this gives existence and uniqueness, and the objective is zero only at w^{HRP}. For $\Sigma = \sum_k \lambda_k \bar{\Sigma}_k$ with λ in the simplex and $d = w - w^{\mathrm{HRP}}$, $d^\top \Sigma d = \sum_k \lambda_k d^\top \bar{\Sigma}_k d \le \max_k d^\top \bar{\Sigma}_k d$, because the tracking-error variance is linear in Σ. □

Problem (7) is a second-order cone program (epigraph form) and solves as fast as (5). The choice of scenarios was made on the validation period only (Appendix A): the variant using the 1-year and 3-month regimes was selected. The hedges-fail stress scenario was rejected at that stage because it lowered validation performance markedly, even though the held-out period later contained exactly such a regime (2022). We report the frozen choice regardless.

5 Experimental Design

Data. We use daily dividend- and split-adjusted closing prices from Yahoo Finance for three universes chosen to span different constraint regimes (Table 1): (i) 20 US-listed multi-asset ETFs covering equities, government and corporate bonds, commodities and real estate (January 2008 – October 2026); (ii) 43 long-listed constituents of India's NIFTY 50 index across 12 sectors (January 2011 – October 2026); (iii) 47 US large-cap stocks across 11 sectors (January 2006 – October 2026). All series are complete over their sample. Survivorship bias affects the stock universes, which consist of current large caps; it affects all methods equally and does not change their ranking, which is our object of study.

Table 1: Universes and constraint sets. Constraint sets were fixed before any backtest.

	Multi-asset ETFs	NIFTY 50 stocks	US large caps
Assets / groups	20 / 4	43 / 12	47 / 11
Sample	2008–2026	2011–2026	2006–2026
Per-asset cap u_i	15%	6%	6%
Group cap $\bar{c}_g$	40%	25%	20%
Group floor $\underline{c}_g$	Equity $\ge$ 20%	–	–
Turnover budget τ (per month)	30%	30%	30%
Validation / test split	2016 | 2017–	2018 | 2019–	2015 | 2016–

Constraints. The constraint sets mirror typical mandates: a single-holding cap of roughly $2.5/N$, sector or asset-class caps, a growth floor of 20% in equities for the multi-asset portfolio, and a monthly two-sided turnover budget of 30%. All weights are long-only and fully invested.

Methods compared. *Unconstrained references*: equal weight (EW), inverse variance (IVP), long-only minimum variance (MinVar), equal risk contribution (ERC) and standard HRP. *Constrained competitors*: minimum variance (MinVar-C) and mean–variance with risk aversion 5 (MVO-C), both solved over the full feasible set; HRP with iterative clipping and renormalisation (HRP-Clip, the common practitioner fix, which ignores turnover); and HRP followed by Euclidean projection onto the full feasible set (HRP-Proj). *Proposed*: the solver-free capacity-aware bisection (CA-HRP-B) and the full method with tracking-error projection (CA-HRP). Covariance-based competitors use the Ledoit–Wolf estimator, except the HRP baselines, which follow the original sample-covariance recipe.

Backtest protocol. At each month end we estimate inputs from the previous $L = 252$ trading days, compute target weights, and trade at the close. Between rebalances weights drift with prices. Turnover is $\|w_t - w_t^-\|_1$ against the drifted weights, and a cost of 10 bps per unit of turnover is

deducted (we also report 0, 25 and 50 bps). Constraint compliance is checked on the target weights at every rebalance.

Design selection and held-out test. An initial version of CA-HRP enforced all constraints inside the bisection followed by a tree-weighted projection. Before finalising the method, we compared eight configurations *on the validation period only* (first part of each sample, Table 1) and selected the tracking-error projection with Ledoit–Wolf covariance (Appendix A). The configuration was then frozen, and we report results both over the full sample and over the untouched test period.

Metrics. Annualised return (CAGR), volatility, Sharpe ratio (zero risk-free rate), maximum drawdown, daily 5% CVaR, annualised turnover, constraint-violation rate (share of rebalances with any breach), realised tracking error to unconstrained HRP ($\mathrm{TE}_{\mathrm{HRP}}$), effective number of holdings $1/\sum_i w_i^2$, and risk concentration, the Herfindahl index of percentage risk contributions scaled by N (1 = equal risk). Sharpe ratio differences are tested with a circular block bootstrap (block of 21 days, 2,000 replications) in the spirit of Ledoit and Wolf [9] and Politis and Romano [16].

6 Results

6.1 Constraint compliance

Table 2 first answers the practical question. Unconstrained HRP breaches the mandate at 100% of rebalances in the multi-asset universe, 48% for NIFTY and 66% for US stocks. The common practitioner fix, HRP-Clip, still breaches constraints at 46%, 8% and 8% of rebalances respectively: alternating clipping and group rescaling can cycle when per-asset and group limits interact, and it ignores turnover entirely. Every optimisation-based method, including both CA-HRP variants, satisfies all constraints at every rebalance, and CA-HRP never needed the turnover fallback.

6.2 Performance where constraints bind

The multi-asset universe is where constraints bite hardest: HRP would hold on average 86% in a single short-term Treasury fund (Figure 2), so the 15% fund cap, the 40% asset-class cap and the 20% equity floor force a very different portfolio. Here the way the constraints are enforced matters a great deal. CA-HRP reaches a Sharpe ratio of 0.908 with 7.4% volatility and a −18.8% maximum drawdown, against 0.764, 9.9% and −22.8% for the Euclidean projection (HRP-Proj) and 0.709 for HRP-Clip. The difference with HRP-Clip is significant ($p = 0.01$); against HRP-Proj and equal weight the p-values are 0.13 and 0.12. Among all constrained methods, only constrained minimum variance has a higher Sharpe ratio (0.935), a difference that is far from significant ($p = 0.34$). CA-HRP's realised tracking error to HRP (6.4%) is the lowest of the HRP-family methods, as its construction intends. The composition explains why. Averaged over all rebalances, HRP-Proj spreads the weight it removes from bonds almost evenly (about 5% in each non-bond fund), whereas CA-HRP keeps the bond budget but shifts it from short-term to intermediate and long Treasuries (IEF 13%, TLT 9%), whose negative correlation with equities hedges the risk it must now take, and holds gold (13%), broad commodities (12%) and the S&P 500 (11%) as its main risky positions.

The solver-free CA-HRP-B performs like the Euclidean projection (0.750) but with four times the turnover. Its bisection handles constraints by pushing excess weight down the tree, so the portfolio concentrates in whichever volatile branch has spare capacity (commodities, Figure 2 and Section 4). This is the central negative finding that motivated the final design: constraints are best enforced *after* the risk allocation, in a risk metric, not inside the bisection.

Table 2: Full-sample out-of-sample results, net of 10 bps costs. CAGR, Vol, MaxDD, CVaR and TE_{HRP} in %; turnover annualised; Viol = share of rebalances breaching any constraint; p = bootstrap p-value of the Sharpe difference with CA-HRP. Unconstrained methods (top block) are shown for reference only. Best constrained Sharpe ratio in bold.

Universe	Method	CAGR	Vol	Sharpe	p	MaxDD	$CVaR_5$	Turn.	Viol	TE_{HRP}	Eff. N
Multi-asset ETFs	EW	8.0	10.9	0.763	0.12	-24.8	-1.63	0.34	0%	10.2	20.0
	IVP	1.9	2.2	0.840	0.70	-9.6	-0.34	0.30	100%	0.5	1.6
	MinVar	2.4	2.4	0.971	0.68	-9.1	-0.36	0.90	100%	1.4	2.5
	ERC	4.8	5.7	0.852	0.47	-17.3	-0.84	0.48	100%	4.5	9.3
	HRP	1.6	2.0	0.833	0.70	-8.8	-0.30	0.63	100%	0.0	1.4
	MinVar-C	6.8	7.3	**0.935**	0.34	-17.5	-1.07	1.21	0%	6.5	8.0
	MVO-C	8.8	12.2	0.753	0.30	-19.1	-1.91	3.16	0%	11.6	7.3
	HRP-Clip	6.3	9.2	0.709	0.01	-22.8	-1.37	2.17	46%	8.4	13.9
	HRP-Proj	7.4	9.9	0.764	0.13	-22.8	-1.49	0.62	0%	9.3	16.1
	CA-HRP-B	7.1	9.8	0.750	0.14	-21.4	-1.47	2.50	0%	9.2	10.7
	CA-HRP	6.6	7.4	0.908	–	-18.8	-1.08	1.25	0%	6.4	8.3
NIFTY 50 stocks	EW	19.7	15.4	1.247	0.07	-36.0	-2.21	0.62	0%	3.7	43.0
	IVP	18.7	14.3	1.272	0.02	-33.4	-2.04	0.85	17%	2.0	37.8
	MinVar	18.0	12.8	1.356	0.89	-26.1	-1.74	2.87	100%	4.7	14.1
	ERC	19.5	14.2	1.326	0.22	-32.7	-2.02	0.81	3%	1.8	39.0
	HRP	19.5	13.7	1.368	0.96	-30.6	-1.94	2.70	48%	0.0	32.4
	MinVar-C	18.6	12.8	**1.401**	0.62	-26.5	-1.76	2.12	0%	3.5	19.8
	MVO-C	20.2	16.2	1.216	0.17	-33.2	-2.35	3.35	0%	6.8	17.4
	HRP-Clip	19.6	13.7	1.369	1.00	-30.6	-1.95	2.66	8%	0.2	32.8
	HRP-Proj	19.6	13.8	1.368	0.97	-30.6	-1.95	2.62	0%	0.2	32.9
	CA-HRP-B	19.5	13.7	1.368	0.68	-30.1	-1.95	2.51	0%	0.9	33.1
	CA-HRP	19.5	13.7	1.369	–	-30.1	-1.94	2.47	0%	0.9	33.1
US large caps	EW	13.3	18.5	0.767	0.59	-47.9	-2.80	0.53	0%	4.8	47.0
	IVP	11.7	16.1	0.768	0.15	-41.1	-2.40	0.73	24%	1.3	37.0
	MinVar	9.3	13.7	0.720	0.39	-33.0	-1.98	3.02	100%	6.2	10.5
	ERC	12.4	16.8	0.781	0.67	-44.7	-2.54	0.68	3%	2.3	42.6
	HRP	11.6	15.7	0.776	0.14	-40.7	-2.35	2.88	66%	0.0	33.9
	MinVar-C	10.7	14.5	0.773	0.66	-35.7	-2.15	2.06	0%	3.6	19.5
	MVO-C	12.7	17.5	0.774	0.82	-43.2	-2.64	3.43	0%	6.9	17.5
	HRP-Clip	11.6	15.8	0.775	0.12	-41.1	-2.37	2.80	8%	0.3	34.6
	HRP-Proj	11.7	15.8	0.775	0.11	-41.2	-2.37	2.76	0%	0.3	34.7
	CA-HRP-B	11.9	15.8	0.790	0.55	-41.9	-2.36	2.65	0%	1.0	34.8
	CA-HRP	11.9	15.7	**0.793**	–	-40.4	-2.35	2.57	0%	0.9	34.7

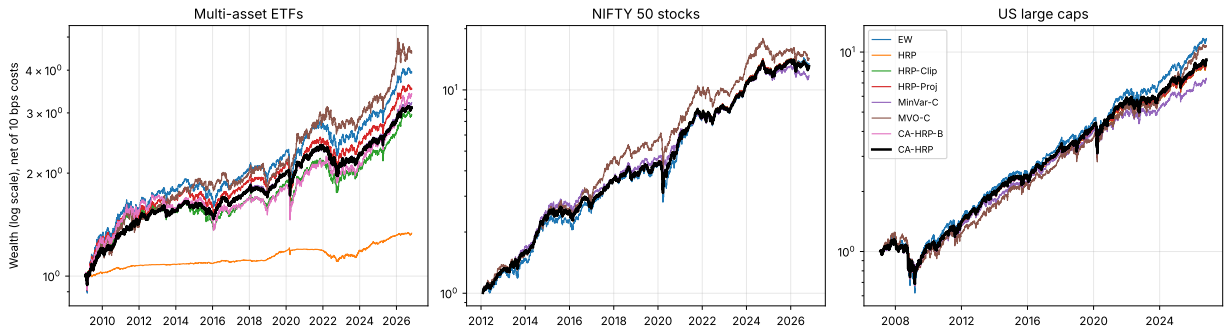


Figure 1: Growth of one unit of wealth (log scale), net of 10 bps costs.

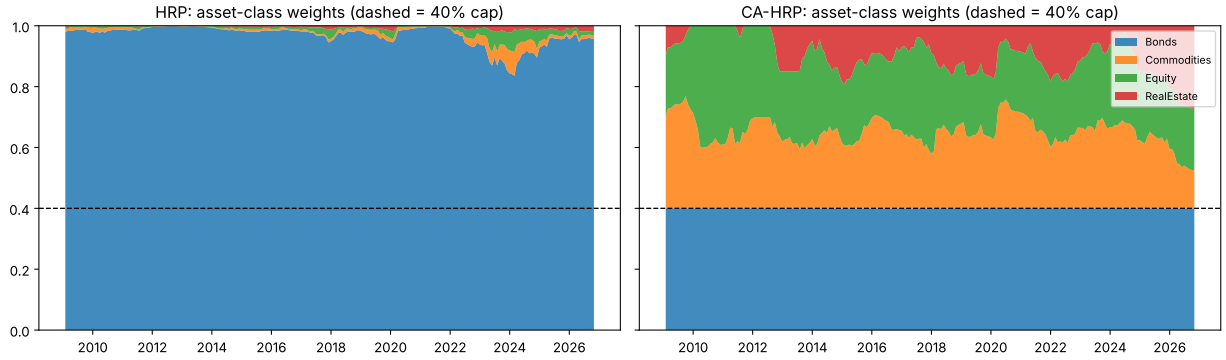


Figure 2: Asset-class weights in the multi-asset universe. Unconstrained HRP (left) is almost entirely in bonds, far above the 40% cap. CA-HRP (right) holds bonds at the cap and allocates the remainder by tracking-error projection.

6.3 Performance where constraints rarely bind

In the two stock universes HRP breaches the limits at about half of the rebalances, but only by small amounts: the realised tracking error between HRP and any of its constrained versions is at most 1%. All HRP-based methods are therefore statistically indistinguishable, which is what Proposition 2(b) predicts as the constraints become slack. CA-HRP posts the highest Sharpe ratio of all eleven methods for US large caps (0.793) and ties HRP for NIFTY (1.369 vs 1.368), where constrained minimum variance is marginally higher (1.401, $p = 0.62$). In other words, CA-HRP costs nothing when HRP is already compliant and helps when it is not.

6.4 Held-out test period

Table 3 repeats the comparison on the period after the validation window, which played no role in choosing the design. The ranking is unchanged. In the multi-asset universe CA-HRP's test Sharpe is 0.929, against 0.805 for HRP-Proj, 0.802 for HRP-Clip, 0.783 for equal weight and 0.977 for constrained minimum variance, with the lowest volatility of any HRP-family method. In stocks CA-HRP again matches HRP.

6.5 Ablation

Table 4 isolates each design choice. In the multi-asset universe the *distance metric* drives almost the whole gain: replacing the full-covariance tracking-error metric by its diagonal lowers the Sharpe ratio from 0.908 to 0.777, and the Euclidean metric to 0.742. Adding capacity-aware bisection before the projection lowers it to 0.861 and nearly doubles turnover; adding a tree-consistency penalty has a similar effect. The choice of covariance estimator, linkage rule and split rule matters

Table 3: Sharpe ratio (net of 10 bps) on the held-out test period, which was never used to choose the CA-HRP design, next to the full sample. Vol and MaxDD (%) on the test period.

	Test-period Sharpe			ETFs, test period		
Method	ETFs	NIFTY	US	Vol	MaxDD	Full-sample SR
EW	0.783	1.066	0.972	10.6	-24.8	0.763
HRP	0.830	1.126	0.909	2.4	-8.8	0.833
MinVar-C	0.977	1.051	0.889	7.4	-17.5	0.935
MVO-C	0.781	0.869	0.976	12.9	-19.1	0.753
HRP-Clip	0.802	1.126	0.909	8.9	-22.8	0.709
HRP-Proj	0.805	1.126	0.910	9.6	-22.8	0.764
CA-HRP-B	0.876	1.122	0.947	9.5	-21.3	0.750
CA-HRP	0.929	1.123	0.941	7.4	-18.8	0.908

little (Sharpe 0.910–0.922), so the method is not sensitive to those HRP details. Removing the turnover cap barely changes performance, showing that the cap is rarely binding for CA-HRP and does not drive its results. In the stock universes no ablation moves the Sharpe ratio by more than 0.04.

Table 4: Ablation of CA-HRP components (full sample, net of 10 bps). Each row changes one component of the full method.

	Sharpe ratio			ETFs		
Variant	ETFs	NIFTY	US	Vol	Turn.	TE_{HRP}
CA-HRP (full)	0.908	1.369	0.793	7.4	1.25	6.4
sample covariance	0.922	1.372	0.781	7.3	1.22	6.4
diagonal metric	0.777	1.366	0.791	9.2	1.14	8.5
Euclidean metric	0.742	1.365	0.789	9.8	1.06	9.1
+ capacity bisection	0.861	1.367	0.796	8.5	2.30	7.7
+ tree-consistency term	0.846	1.370	0.794	7.9	2.27	7.1
no turnover cap	0.910	1.369	0.793	7.3	1.26	6.4
Ward linkage	0.910	1.330	0.759	7.4	1.30	6.4
dendrogram splits	0.918	1.392	0.778	7.4	1.19	6.5

6.6 Robustness

Costs and estimation window. Table 5 shows the multi-asset advantage of CA-HRP over the other HRP-family methods survives 50 bps costs (0.840 vs 0.739 for HRP-Proj) and both shorter (126-day) and longer (504-day) estimation windows, where CA-HRP improves on HRP-Proj by 0.14 and 0.19 Sharpe points respectively. Constrained minimum variance remains a strong competitor throughout.

Stability of weights. We re-estimate every method on 30 block-bootstrapped versions of the estimation window at eight dates per universe and measure how much the weights move. CA-HRP is far more stable than mean–variance (instability 0.16–0.22 vs 0.70–0.78) and more stable than constrained minimum variance in every universe (e.g. 0.218 vs 0.324 for NIFTY). In the multi-asset universe HRP-Proj is the most stable, because it is anchored near equal weight; its stability comes at the price of a portfolio that tracks HRP poorly and has higher volatility.

Table 5: Robustness. Left: Sharpe ratio in the multi-asset universe at 50 bps costs and with 126- and 504-day estimation windows (all universes). Right: weight instability, the mean L_1 distance of bootstrapped weights from their average (lower = more stable).

	ETFs	Window 126 days			Window 504 days			Weight instability		
Method	50 bps	ETFs	NIFTY	US	ETFs	NIFTY	US	ETFs	NIFTY	US
EW	0.750	0.535	1.197	0.794	0.701	1.196	0.775	–	–	–
HRP	0.705	0.622	1.288	0.786	0.932	1.287	0.767	0.068	0.230	0.227
MinVar-C	0.868	0.703	1.392	0.801	0.936	1.341	0.730	0.185	0.324	0.318
MVO-C	0.650	0.660	1.142	0.755	0.796	1.172	0.760	0.754	0.698	0.781
HRP-Clip	0.614	0.479	1.293	0.786	0.800	1.288	0.767	0.209	0.222	0.214
HRP-Proj	0.739	0.535	1.293	0.792	0.714	1.288	0.766	0.037	0.220	0.217
CA-HRP	0.840	0.676	1.301	0.803	0.908	1.276	0.766	0.158	0.218	0.211

Stress periods. Table 6 reports cumulative returns in three crises. In the COVID crash CA-HRP lost 6.9% in the multi-asset universe, against 14.0% for HRP-Proj, 13.7% for HRP-Clip and 14.9% for equal weight, and it was flat in the GFC window. The 2022 rates shock is the exception: CA-HRP's hedge tilt towards intermediate and long Treasuries, which protected it in 2020, lost 17.2% when bonds and equities fell together, slightly more than HRP-Proj (−15.3%). This illustrates the main limitation of any covariance-based hedge: it relies on correlations that can break down.

Table 6: Cumulative return (%) in stress windows: GFC = Sep 2008–Mar 2009, COVID = 15 Feb–30 Apr 2020, Rates = Jan–Oct 2022.

	ETFs			NIFTY	US		
Method	GFC	COVID	Rates	COVID	GFC	COVID	Rates
EW	-1.6	-14.9	-16.9	-16.8	-30.4	-13.5	-5.1
HRP	0.6	-0.3	-6.6	-9.6	-24.8	-10.0	-5.9
MinVar-C	0.5	-7.6	-16.0	-4.4	-20.3	-10.7	-4.4
MVO-C	0.8	-4.5	-8.2	-13.8	-30.1	-9.3	-0.9
HRP-Clip	0.4	-13.7	-17.4	-9.6	-24.8	-10.0	-6.3
HRP-Proj	-1.0	-14.0	-15.3	-9.6	-25.0	-10.0	-6.3
CA-HRP-B	-2.2	-11.2	-11.3	-8.4	-25.8	-9.0	-5.3
CA-HRP	0.3	-6.9	-17.2	-8.4	-24.7	-9.5	-5.4

Tightening the constraints. Finally, we tighten the per-asset cap from $4/N$ to $1.25/N$ (with group caps scaled accordingly) and measure, at twelve dates per universe, how far each method's portfolio is from HRP in ex-ante tracking-error terms (Figure 3). Settings that are mathematically infeasible (caps of $1.5/N$ and $1.25/N$ in the multi-asset universe) are detected by a linear feasibility check and skipped; HRP-Clip silently returns infeasible weights in such cases, and even at caps of $2.5/N$ and looser it still breaches limits by 1.1–1.9 percentage points on average in the multi-asset universe. In that universe CA-HRP has the smallest distortion at every cap (7.1–9.2% versus 10.5–11.9% for HRP-Proj) and the lowest volatility. In the stock universes CA-HRP has a small constant floor of about 0.8% because its HRP target uses shrinkage rather than the sample covariance; at the tightest cap ($1.25/N$) it distorts HRP the least in both stock universes (2.2% vs 2.4–2.5% for NIFTY, 2.8% vs 3.2–3.3% for US), and already at $1.5/N$ for US stocks.

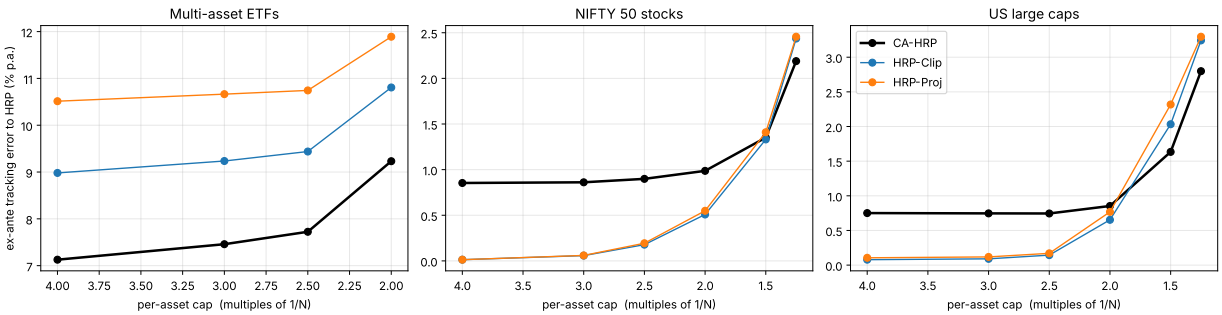


Figure 3: Ex-ante tracking error to sample-covariance HRP as the per-asset cap tightens (right = tighter). Infeasible settings are omitted.

7 Extended Evaluation

After the main study was complete and the method frozen, we ran a second, broader round of tests designed to probe its weaknesses rather than its strengths. All constraint sets and the CA-HRP-R scenario choice were fixed before any of these results were computed.

7.1 External validation on three new universes

We added three universes that played no role in any design decision: six Indian exchange-traded funds (Nifty 50, Bank, Junior, PSU Bank, Nasdaq-100 and Gold BeES; April 2011 – October 2026) under a balanced-fund mandate (gold $\leq 25\%$, international equity $\leq 35\%$, Indian equity between 40% and 80%, single fund $\leq 35\%$); the ten US sector SPDR funds (June 2000 – October 2026), grouped as cyclical or defensive with defensive sectors capped at 50% and single sectors at 20%; and nineteen single-country equity funds (October 2004 – October 2026) grouped into four regions, each capped at 40% with single countries at 10%. Yahoo Finance records two short-lived unadjusted unit splits in the Indian funds (December 2019 and June 2021), visible as -90% / $+900\%$ return pairs; we detect such pairs automatically and rescale the affected prices by the implied split ratio (10 or 100). No other universe is affected.

Table 7 reports the results. Unconstrained HRP would breach these mandates at 94%, 92% and 97% of rebalances respectively, and HRP-Clip still breaches them at 1%, 14% and 9%. CA-HRP is fully compliant. Its Sharpe ratio is above HRP-Proj in the Indian (1.276 vs 1.257) and sector (0.628 vs 0.619) universes and slightly below it for countries (0.428 vs 0.436), and in every universe it has the lowest volatility of the three HRP fixes. Constrained minimum variance (Indian ETFs) and mean–variance (sectors) post the best Sharpe ratios. The external evidence therefore confirms the pattern of the main study: CA-HRP is a reliable way to make HRP compliant, with gains that scale with how hard the constraints bind, but it does not dominate optimisation-based methods.

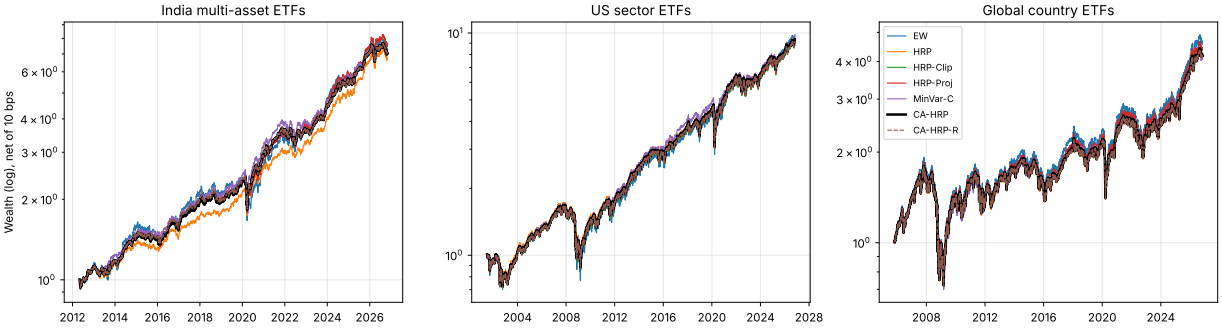


Figure 4: Growth of one unit of wealth (log scale), net of 10 bps, in the three external universes.

Table 7: External validation on three universes that were added after the method was frozen (net of 10 bps). Vol and MaxDD in %; Viol = share of rebalances breaching a constraint. Best constrained Sharpe ratio per universe in bold.

	India ETFs				US sectors				Countries			
Method	Sharpe	Vol	MaxDD	Viol	Sharpe	Vol	MaxDD	Viol	Sharpe	Vol	MaxDD	Viol
EW	1.086	13.7	-31.1	0%	0.586	18.3	-55.2	0%	0.432	22.8	-61.7	100%
HRP	1.336	10.5	-21.5	94%	0.641	15.7	-45.9	92%	0.427	21.1	-59.5	97%
MinVar-C	**1.321**	10.9	-24.9	0%	0.653	15.7	-48.1	0%	0.435	20.1	-60.8	0%
MVO-C	1.052	14.5	-28.1	0%	**0.658**	17.0	-47.8	0%	0.386	22.1	-64.8	0%
HRP-Clip	1.269	11.8	-25.1	1%	0.613	16.6	-49.1	14%	0.435	21.5	-60.4	9%
HRP-Proj	1.257	11.9	-25.6	0%	0.619	16.5	-48.6	0%	**0.436**	21.5	-60.1	0%
CA-HRP-B	1.255	11.9	-25.2	0%	0.603	16.7	-48.9	0%	0.412	21.6	-61.0	0%
CA-HRP	1.276	11.3	-24.9	0%	0.628	16.2	-47.6	0%	0.428	21.2	-60.0	0%
CA-HRP-R	1.275	11.3	-25.6	0%	0.625	16.2	-47.4	0%	0.426	21.2	-60.1	0%

7.2 Statistical precision across six universes

Table 8 and Figure 5 give bootstrap confidence intervals for the Sharpe-ratio difference between CA-HRP and its competitors. Only one interval excludes zero: CA-HRP versus HRP-Clip in the multi-asset ETF universe (+0.20, 95% interval $[+0.03, +0.35]$). The point estimate against HRP-Proj is positive in five of six universes and is large only for multi-asset ETFs (+0.145, $[-0.04, +0.32]$). Against constrained minimum variance it is negative in five of six universes but never significant. We therefore claim reliability and lower risk, not superior returns.

Table 8: Annualised Sharpe-ratio difference of CA-HRP minus each competitor, with 95% circular-block-bootstrap intervals (21-day blocks, 2,000 draws). Intervals excluding zero in bold.

Universe	HRP-Proj	HRP-Clip	MinVar-C	EW	CA-HRP-R
ETFs	+0.145 [-0.04, +0.32]	**+0.199 [+0.03, +0.35]**	-0.027 [-0.08, +0.03]	+0.146 [-0.04, +0.31]	+0.025 [-0.03, +0.08]
NIFTY	+0.001 [-0.03, +0.03]	+0.000 [-0.03, +0.03]	-0.032 [-0.15, +0.09]	+0.122 [-0.01, +0.25]	+0.002 [-0.00, +0.00]
US	+0.018 [-0.00, +0.04]	+0.018 [-0.00, +0.04]	+0.021 [-0.07, +0.11]	+0.027 [-0.08, +0.12]	+0.000 [-0.00, +0.00]
India ETFs	+0.019 [-0.07, +0.11]	+0.007 [-0.06, +0.07]	-0.045 [-0.14, +0.04]	+0.189 [-0.01, +0.38]	+0.001 [-0.03, +0.03]
US sectors	+0.010 [-0.01, +0.03]	+0.016 [-0.01, +0.04]	-0.024 [-0.08, +0.02]	+0.043 [-0.02, +0.11]	+0.003 [-0.00, +0.01]
Countries	-0.008 [-0.02, +0.01]	-0.007 [-0.02, +0.01]	-0.007 [-0.05, +0.03]	-0.004 [-0.04, +0.03]	+0.002 [-0.00, +0.01]

7.3 Does regime-robustness help?

CA-HRP-R (Section 4.5) was designed to address the 2022 weakness. It did not help. Its Sharpe ratio is lower than CA-HRP's in all six universes, though never by more than 0.025, and it trades more (2.09 vs 1.25 annual turnover in the multi-asset universe). In the 2022 rates shock it lost 18.1% in the multi-asset universe against 17.2% for CA-HRP. The reason is visible in the validation study (Table 19): the only scenario that would have anticipated 2022, the hedges-fail stress covariance, made the portfolio much more conservative in normal times and was rejected on validation data. Robustness to a regime that the estimation window has never seen cannot be bought cheaply with covariance scenarios alone; we report this as a negative result.

7.4 Rebalancing frequency

Table 9 repeats the comparison with weekly and quarterly rebalancing (turnover cap of 30% per rebalance in both cases). CA-HRP satisfies all constraints at all frequencies, while HRP-Clip breaches them at 9% (weekly), 14% (monthly) and 27% (quarterly) of rebalances on average across universes. Over the 18 frequency–universe combinations, CA-HRP beats HRP-Proj in 12

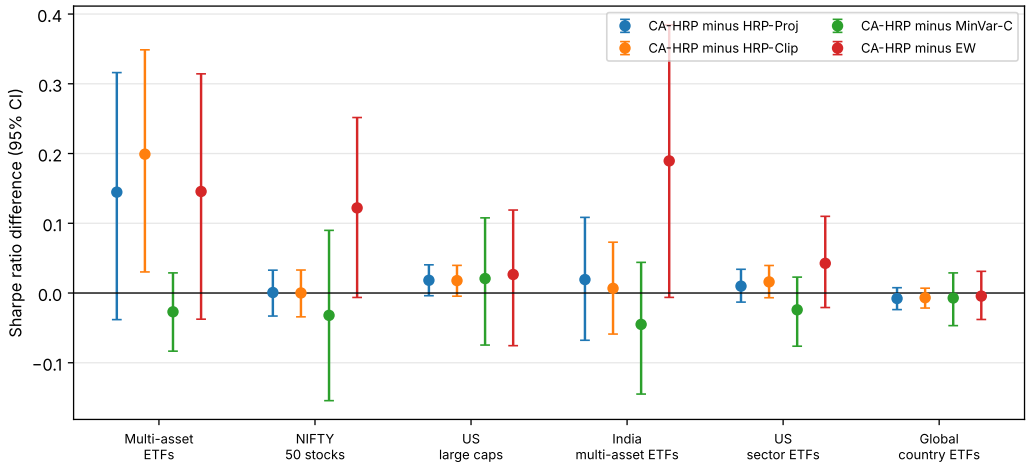


Figure 5: Sharpe-ratio difference of CA-HRP minus each competitor, with 95% block-bootstrap intervals, for all six universes.

and is the best HRP-based method in 10, while constrained minimum variance beats CA-HRP in 16. The multi-asset ranking (CA-HRP clearly ahead of the other HRP fixes, slightly behind minimum variance) holds at every frequency.

Table 9: Sharpe ratio by rebalancing frequency (W = weekly, M = monthly, Q = quarterly), net of 10 bps, all six universes. The first column also gives HRP-Clip's share of rebalances with a constraint breach, pooled over universes (CA-HRP: 0% at every frequency).

Freq.	Universe	EW	MinVar-C	HRP-Clip	HRP-Proj	CA-HRP	CA-HRP-R
W (Clip 9%)	ETFs	0.714	0.888	0.608	0.719	**0.844**	0.819
	NIFTY	1.268	1.414	**1.329**	1.328	1.326	1.325
	US	0.779	0.781	**0.765**	0.763	0.763	0.763
	India ETFs	1.066	1.331	1.258	1.241	**1.269**	1.261
	US sectors	0.588	0.655	0.600	0.609	**0.612**	0.609
	Countries	0.434	0.438	0.404	**0.407**	0.397	0.396
M (Clip 14%)	ETFs	0.763	0.935	0.709	0.764	**0.908**	0.884
	NIFTY	1.247	1.401	1.369	1.368	**1.369**	1.367
	US	0.767	0.773	0.775	0.775	**0.793**	0.793
	India ETFs	1.086	1.321	1.269	1.257	**1.276**	1.275
	US sectors	0.586	0.653	0.613	0.619	**0.628**	0.625
	Countries	0.432	0.435	0.435	**0.436**	0.428	0.426
Q (Clip 27%)	ETFs	0.807	0.920	0.719	0.800	**0.899**	0.873
	NIFTY	1.233	1.375	1.357	1.354	1.361	**1.361**
	US	0.787	0.744	**0.812**	0.808	0.807	0.808
	India ETFs	1.107	1.317	1.279	1.261	1.295	**1.299**
	US sectors	0.593	0.643	0.620	0.625	**0.634**	0.631
	Countries	0.415	0.416	0.421	**0.423**	0.409	0.409

7.5 Which constraints matter?

Table 10 imposes the constraint families one at a time. In the multi-asset universe almost all of CA-HRP's advantage comes from the *group* constraints: with group caps and floors alone its Sharpe ratio is 0.972 against 0.745 for HRP-Proj and 0.684 for HRP-Clip, whereas with per-asset caps

alone the gap is much smaller (0.853 vs 0.802). This is consistent with Proposition 2(d): group caps remove a large block of correlated weight at once, which is exactly where choosing replacements by covariance rather than uniformly pays off. With per-asset caps alone in the Indian ETF universe, CA-HRP is slightly worse than the simple fixes (1.300 vs 1.312–1.314). HRP-Clip never breaches per-asset caps on their own; its breaches come from group caps and floors (36% of rebalances with groups alone in the multi-asset universe) and, in the sector universe, only from the turnover budget that it ignores.

Table 10: Which constraints matter? Sharpe ratio (net of 10 bps) when only some constraint families are imposed. Best HRP-family method per row in bold.

Universe	Constraints	HRP-Clip	HRP-Proj	CA-HRP
ETFs	box only	0.777	0.802	**0.853**
	groups only	0.684	0.745	**0.972**
	box + groups	0.709	0.764	**0.910**
	all (+ turnover)	0.709	0.764	**0.908**
India ETFs	box only	**1.314**	1.312	1.300
	groups only	1.269	1.257	**1.273**
	box + groups	1.269	1.257	**1.273**
	all (+ turnover)	1.269	1.257	**1.276**
US sectors	box only	0.614	0.612	**0.628**
	groups only	0.626	0.635	**0.639**
	box + groups	0.613	0.620	**0.628**
	all (+ turnover)	0.613	0.619	**0.628**
US	box only	0.775	0.774	**0.792**
	groups only	0.777	0.776	**0.795**
	box + groups	0.775	0.774	**0.793**
	all (+ turnover)	0.775	0.775	**0.793**

7.6 Monte Carlo evidence with a known covariance

Backtests mix estimation error with luck. To isolate the former we simulate 40 assets in five correlated clusters, one of which is a low-volatility, negatively correlated "bond" block, so that HRP on the *true* covariance puts 90% in that block and breaches a per-asset cap of $2.5/N$ and group caps of 30%. Groups coincide with clusters except for two assets per cluster, so constraints cut across the tree. Every asset has an annualised Sharpe ratio of 0.4. For each of 200 replications per cell we draw $T \in \{126, 252, 504\}$ daily returns (Gaussian or Student-t_5), estimate each method, and evaluate the resulting portfolio under the true parameters.

Table 11 and Figure 6 show the Student-t case (the Gaussian case is almost identical). CA-HRP has the highest true Sharpe ratio of all methods in every cell (0.809–0.820 under fat tails, 0.813–0.822 under normality). Among the HRP-based methods it has the lowest true tracking error to the oracle HRP portfolio (6.0–6.3%, against 7.3–7.4% for HRP-Clip and 8.2% for HRP-Proj), on par with constrained minimum variance (within 0.05 percentage points). Its true volatility also matches constrained minimum variance (6.4–6.7%), while HRP-Proj is 8.5% and constrained mean–variance 13.7–14.3%. The cross-replication standard deviation of the true Sharpe ratio is below 0.01, so these differences are precisely estimated. In this controlled setting, which isolates the constraint-handling problem the paper is about, CA-HRP is the best method.

Table 11: Monte Carlo study with a known covariance (40 assets, 5 clusters, 200 replications per cell). True Sharpe ratio and true volatility (%) of the estimated portfolio, and true tracking error (%) to HRP computed on the true covariance. Best value per column among constrained methods in bold.

	True Sharpe ratio			True volatility			True TE to true HRP		
Method (Student-t_5)	T=126	252	504	126	252	504	126	252	504
EW	0.722	0.722	0.722	12.62	12.62	12.62	12.49	12.49	12.49
MVO-C	0.680	0.683	0.684	14.00	13.80	13.86	13.85	13.65	13.71
MinVar-C	0.808	0.809	0.811	**6.63**	**6.52**	6.44	**6.28**	6.17	6.09
HRP-Clip	0.806	0.806	0.808	7.74	7.74	7.72	7.39	7.39	7.37
HRP-Proj	0.804	0.804	0.804	8.53	8.54	8.52	8.18	8.20	8.18
CA-HRP	**0.809**	**0.813**	**0.820**	6.66	6.52	**6.43**	6.32	**6.17**	**6.04**
CA-HRP-R	0.808	0.808	0.811	6.73	6.64	6.57	6.39	6.31	6.22

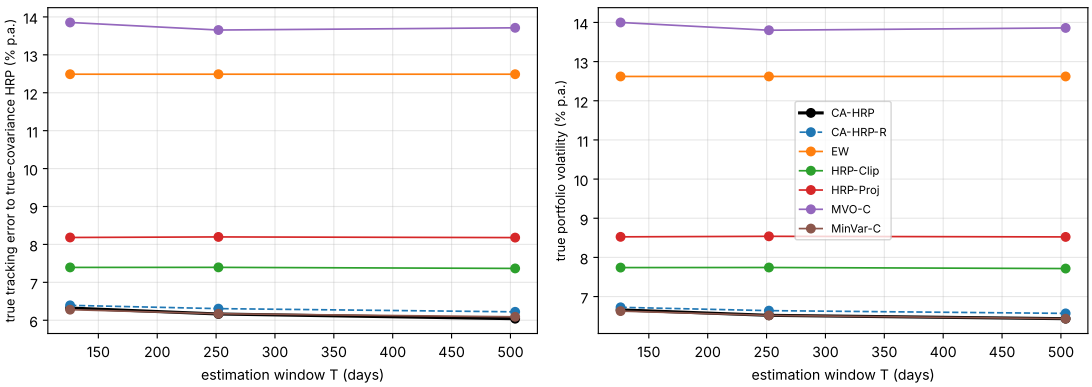


Figure 6: Monte Carlo (Student-t_5): true tracking error to the oracle HRP portfolio and true volatility as a function of the estimation window.

7.7 Computational cost

Figure 7 reports the median time per rebalance for random universes of $N = 10$ to 400 assets with ten groups and a turnover cap. CA-HRP takes 17 ms at $N = 50$ and 0.27 s at $N = 400$, the same order as constrained minimum variance (0.25 s). The solver-free CA-HRP-B takes 0.15 s at $N = 400$ and the regime-robust CA-HRP-R 1.4 s, so the robust variant is the only one whose cost grows noticeably. All are negligible for monthly rebalancing.

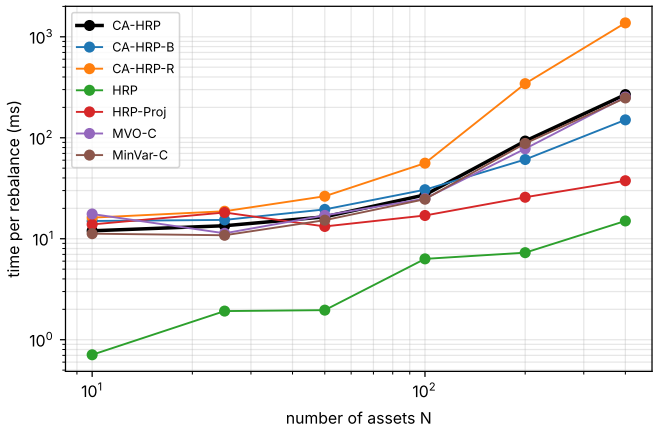


Figure 7: Median computation time per rebalance against the number of assets (log–log).

8 Explaining, Stress-Testing and Verifying the Method

This section adds a third round of analysis: tools that explain *what* compliance costs and *why* CA-HRP is stable, a stress test of the evidence against the objection that the constraint sets were hand-picked, stricter statistics, and an independent verification of the implementation.

8.1 What does compliance cost, and which rule is responsible?

Using Proposition 4 and the Shapley split we decompose, on every one of the 1,363 CA-HRP rebalances, the tracking error that compliance forces away from HRP (Table 12, Figure 8). Two checks hold on every date: re-solving the attribution problem from scratch reproduces the stored backtest weights to within 2.4×10^{-10}, and the turnover remainder in (6) is below 10^{-8} whenever the turnover cap is slack, as the identity requires.

The price of compliance differs sharply across mandates. In the multi-asset universe CA-HRP must sit 5.0% (ex-ante tracking error, p.a.) away from HRP on average, and 14.6% at the peak in 2009, when HRP wanted almost the entire portfolio in short-term Treasuries. In the stock universes compliance is nearly free (0.1% on average) and costs nothing at all on 59% (NIFTY) and 41% (US) of rebalance dates. Wherever compliance is expensive, *group* limits are responsible for most of it: 67% of the cost for multi-asset ETFs, 73% for US sectors, 77% for countries and 82% for Indian ETFs. Only in the NIFTY universe, where sector caps are loose relative to the index, do single-stock caps dominate (97%). The turnover cap binds on at most 12% of dates; its 20% Shapley share in the multi-asset universe reflects that, on its own, a turnover budget would also stop the portfolio from moving towards a target that the other limits have already pushed far away, which the Shapley value credits to it. This attribution is, to our knowledge, new for hierarchical allocation and gives a portfolio manager a direct answer to "which rule of my mandate is costing me the most HRP-ness?".

Table 12: Price of compliance for CA-HRP: ex-ante tracking error to HRP (% p.a.) needed to satisfy the mandate, share of rebalances where compliance is free, share where the turnover cap binds, and the Shapley split of the cost across constraint families.

	TE to HRP		Free	Turnover	Shapley share		
Universe	mean	max	dates	binds	asset caps	groups	turnover
ETFs	5.0	14.6	0%	1%	14%	67%	20%
NIFTY	0.1	0.5	59%	7%	97%	1%	2%
US	0.1	0.9	41%	6%	46%	51%	3%
India ETFs	2.8	6.8	3%	3%	13%	82%	5%
US sectors	1.0	3.8	10%	12%	24%	73%	3%
Countries	0.7	2.2	2%	11%	22%	77%	1%

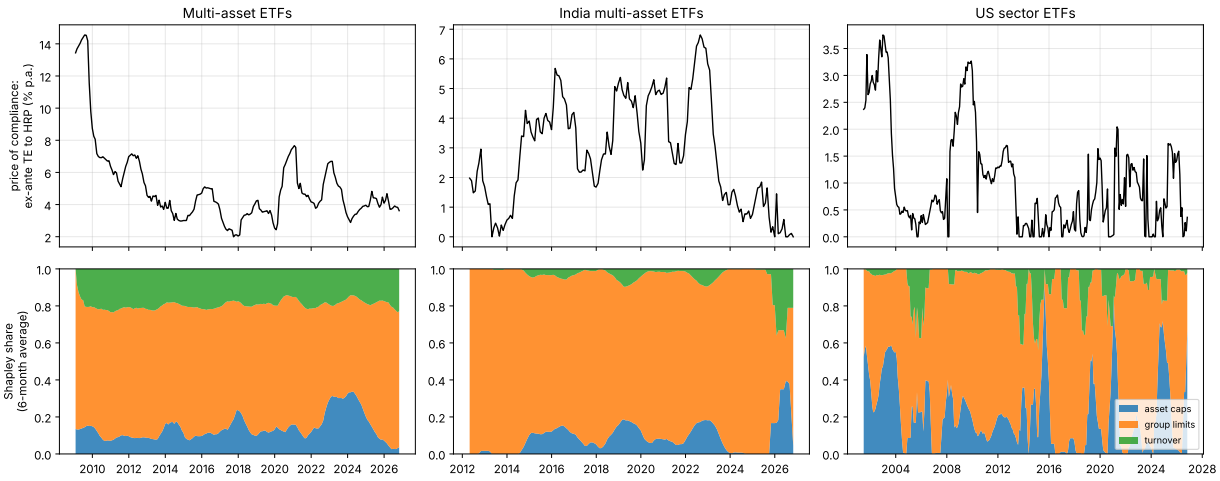


Figure 8: Top: price of compliance (ex-ante tracking error of CA-HRP to HRP). Bottom: Shapley share of each constraint family (6-month moving average).

8.2 Noise damping: CA-HRP shrinks estimation noise, clipping amplifies it

To test Proposition 3 empirically we bootstrap the estimation window 40 times at 10 dates per universe, compute HRP on each resample, and push each target through the CA-HRP projection and through HRP-Clip. For all $6 \times 10 \times 780$ pairs of resamples we compare the distance between the two portfolios with the distance between the two targets (Table 13). The CA-HRP projection never increased the distance beyond solver tolerance (largest ratio $1 + 10^{-6}$), as the proposition guarantees, and on average it shrank it, to 0.48 of its original size for multi-asset ETFs and to 0.74–0.97 elsewhere. Clipping behaves very differently. In the multi-asset universe it amplified noise on average (ratio 2.08) and by a factor of 10.5 for the worst pair, and it pushed half of all pairs further apart; in the Indian ETF and sector universes it amplified 43% and 25% of pairs. This is the mechanism behind HRP-Clip's high turnover in Table 2: small changes in the estimate produce large changes in the clipped portfolio.

Table 13: Noise damping. Ratio of the dispersion of bootstrapped portfolios to the dispersion of the bootstrapped HRP targets (1 = noise passed through unchanged, <1 = damped, >1 = amplified), averaged over 10 dates × 780 bootstrap pairs, and the worst pair. Measured in the risk metric $\bar{\Sigma}$; "share amplified" is the fraction of pairs that clipping pushes further apart in the Euclidean norm.

	CA-HRP projection		HRP-Clip		Share
Universe	mean	worst	mean	worst	amplified
ETFs	0.48	0.99	2.08	10.49	50%
NIFTY	0.97	1.00	0.96	1.06	3%
US	0.97	1.00	0.95	1.17	4%
India ETFs	0.74	1.00	0.90	1.99	43%
US sectors	0.85	1.00	0.84	1.75	25%
Countries	0.82	1.00	0.83	1.63	14%

8.3 Random mandates and a placebo test

The constraint sets in Sections 6–7 were fixed in advance, but they are still only nine mandates. To remove any suspicion that they favour CA-HRP, we drew 160 random mandates, 40 in each of four universes (multi-asset ETFs, NIFTY, US large caps and countries). Each draw picks a random subset of between half and all of the assets, a random start date in the first third of the sample, a per-asset cap between $1.5/n$ and $4/n$, a group cap between $1.2/G$ and $2.5/G$, and a 30% turnover budget; infeasible draws are rejected by a linear feasibility check. As a placebo, every second draw replaces the true sector, asset-class or region labels by random labels into three to six groups, which breaks the link between groups and correlation structure. The design and random seeds were fixed before the study was run.

Table 14 and Figure 9 summarise the results. CA-HRP and the other optimisation-based methods satisfied every constraint in every mandate, while HRP-Clip breached them at 4% of rebalances. CA-HRP had volatility no higher than HRP-Proj in 85% of the mandates. On Sharpe ratio, however, the picture is balanced: CA-HRP beat HRP-Proj in 55% of mandates (mean difference +0.014, median +0.002), HRP-Clip in 56% and constrained minimum variance in 46%. The advantage is concentrated in the multi-asset universe, where CA-HRP beat HRP-Proj in 90% of mandates with true groups, and it reverses for country funds, where it won only 25–30% of the time. Scrambling the group labels halves the multi-asset advantage (+0.076 to +0.035), consistent with the mechanism of Proposition 2(d): when groups follow the correlation structure, choosing replacement assets by covariance matters more. In the US universe the placebo did not weaken the effect, so this evidence is suggestive rather than conclusive. The gain is largest when HRP breaches the mandate at more than 75% of rebalances (+0.021, 60% win rate), but it is not monotone in

how often the mandate binds. Overall the random mandates support the paper's central, narrower claim: CA-HRP is a reliable and lower-risk way to make HRP compliant, with a return advantage that is specific to strongly binding, correlation-aligned group constraints.

Table 14: Random mandates (40 per universe, half with scrambled group labels). Share of mandates in which CA-HRP has a higher Sharpe ratio than each competitor, mean Sharpe difference with HRP-Proj, and share in which CA-HRP's volatility is not higher than HRP-Proj's.

Universe	Groups	n	CA-HRP wins vs			ΔSR	Vol ≤
			Proj	Clip	MinVar-C	vs Proj	Proj
Countries	true	20	30%	25%	30%	-0.002	90%
Countries	scrambled	20	25%	30%	25%	-0.005	90%
ETFs	true	20	90%	95%	10%	+0.076	100%
ETFs	scrambled	20	75%	95%	30%	+0.035	100%
NIFTY	true	20	45%	40%	70%	+0.001	70%
NIFTY	scrambled	20	45%	40%	55%	-0.002	40%
US	true	20	60%	55%	65%	+0.002	95%
US	scrambled	20	70%	70%	85%	+0.009	95%
All	all	160	55%	56%	46%	+0.014	85%
By how often HRP breaches the mandate:							
HRP breach share (-0.01, 0.25]		21	48%			+0.004	
HRP breach share (0.25, 0.5]		17	59%			+0.007	
HRP breach share (0.5, 0.75]		21	33%			-0.004	
HRP breach share (0.75, 1.0]		101	60%			+0.021	

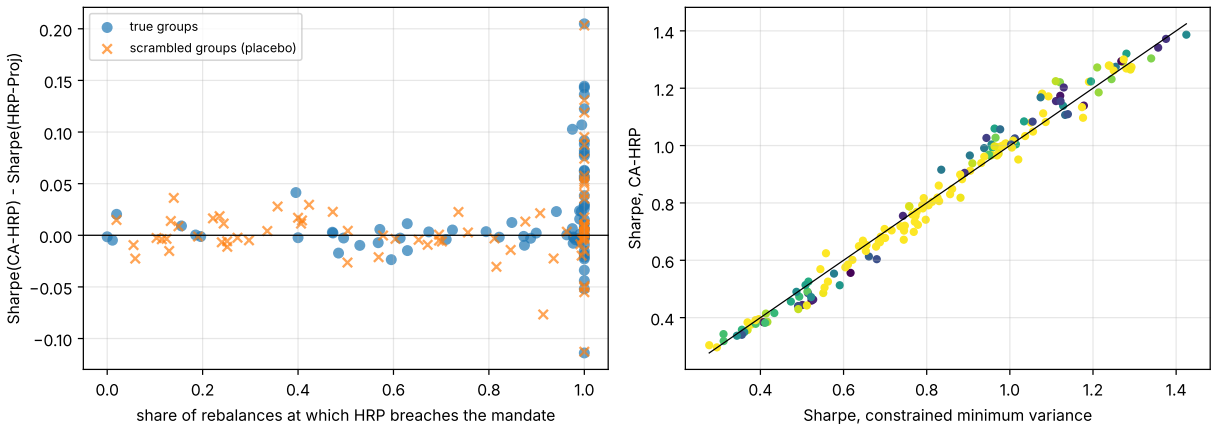


Figure 9: Random mandates. Left: Sharpe-ratio gain of CA-HRP over HRP-Proj against how often HRP breaches the mandate (circles: true groups; crosses: scrambled-group placebo). Right: CA-HRP against constrained minimum variance (colour: HRP breach share).

8.4 Forward-looking risk from option-implied volatility

All risk inputs so far are backward-looking. Option prices, by contrast, embed the market's forward-looking view of volatility, which often moves before realised volatility does. We therefore test a variant, CA-HRP-IV, that keeps the historical (Ledoit–Wolf) correlations but rescales each asset's volatility by the implied-to-realised ratio of the volatility index of its asset class:

$$\sigma_i^{\mathrm{IV}}(d) = \sigma_i^{\mathrm{hist}}(d)\,\frac{\mathrm{IV}_{k(i)}(d^-)}{\mathrm{RV}_{k(i)}(d^-)}, \tag{8}$$

where IV_k is the index level on the last trading day strictly before the rebalance date d and RV_k the realised volatility of its underlying over the 252 preceding days. We use VIX (US and international equity and real estate), VXN (Nasdaq-100), GVZ (gold and silver), OVX (oil and broad commodities), MOVE (Treasuries and investment-grade credit, with realised volatility measured on daily 10-year yield changes in basis points) and India VIX (Indian equity). Assets without a suitable index (high-yield and emerging-market bonds) receive the median ratio of the available indices, so that the variance risk premium does not make them look artificially safe. The resulting covariance replaces the historical one in both the HRP allocation and the projection metric. The one-day lag avoids look-ahead across time zones, and the mapping was fixed before any backtest. To attribute any effect to the *information* rather than the method, the same covariance is also given to constrained minimum variance (MinVar-C-IV).

Proposition 6 (Scale invariance)**.** *HRP weights and the CA-HRP projection are unchanged when every volatility is multiplied by the same factor $\kappa > 0$.*

Proof. Correlations, and therefore the tree, are unchanged. Every cluster variance in (2) scales by κ^2, so the split fractions do not change. The projection metric in (5) is normalised by its trace, which removes κ^2. □

Proposition 6 implies that option information can only change CA-HRP through the *relative* implied/realised ratios across asset classes, so a single index such as VIX applied to a one-asset-class universe would have no effect; this is why we test it on the two multi-asset universes. Both properties, together with the absence of look-ahead in (8), are covered by unit tests, and with the ratios set to one the IV code path reproduces CA-HRP exactly.

The ratios vary widely (Figure 10): at month-ends they peaked at 2.9 for VIX (February 2020), 3.3 for OVX and 3.1 for India VIX (March 2020), and during 2022 MOVE was on average 25% above realised Treasury volatility (monthly range 0.97–1.55). Table 15 shows the outcome, which is mixed. Option information did not change risk-adjusted returns: CA-HRP-IV has a Sharpe ratio of 0.907 against 0.908 for CA-HRP in the multi-asset universe and 1.280 against 1.276 for Indian ETFs, with intervals centred on zero. It did improve crash behaviour in the US multi-asset universe, cutting the COVID-crash loss from 6.9% to 5.5% and the 2022 loss from 17.2% to 16.4%, but it worsened the COVID loss for Indian ETFs (7.9% against 5.3%). It also roughly doubled turnover (2.10 against 1.25 and 1.91 against 0.98 per year), because implied/realised ratios move from month to month. Constrained minimum variance benefited somewhat more from the same information (Sharpe 1.355 against 1.321 for Indian ETFs). We conclude that implied volatility is a usable, look-ahead-free input to CA-HRP whose benefit is limited to some crisis episodes; smoothing the ratios to contain turnover is a natural extension, which we leave for future work rather than tune on these data.

Table 15: Forward-looking risk with option-implied volatility (net of 10 bps). CA-HRP-IV and MinVar-C-IV use historical correlations and volatilities rescaled by implied/realised ratios of VIX, VXN, GVZ, OVX, MOVE and India VIX. Vol, MaxDD and stress returns in %; ΔSR = Sharpe of CA-HRP-IV minus that method, with 95% block-bootstrap interval.

Universe	Method	Sharpe	Test SR	Vol	MaxDD	Turn.	GFC	COVID	2022	ΔSR [95% CI]
Multi-asset ETFs	HRP-Proj	0.764	0.805	9.9	-22.8	0.62	-1.0	-14.0	-15.3	+0.143 [-0.07, +0.34]
	MinVar-C	0.935	0.977	7.3	-17.5	1.21	0.5	-7.6	-16.0	-0.028 [-0.13, +0.07]
	MinVar-C-IV	0.932	0.959	7.3	-17.1	1.96	0.1	-5.7	-15.2	-0.025 [-0.08, +0.03]
	CA-HRP	0.908	0.929	7.4	-18.8	1.25	0.3	-6.9	-17.2	-0.001 [-0.09, +0.08]
	CA-HRP-IV	0.907	0.913	7.3	-18.3	2.10	0.1	-5.5	-16.4	–
India ETFs	HRP-Proj	1.257	–	11.9	-25.6	0.69	–	-7.1	2.7	+0.023 [-0.11, +0.15]
	MinVar-C	1.321	–	10.9	-24.9	0.79	–	-4.0	-1.8	-0.041 [-0.16, +0.07]
	MinVar-C-IV	1.355	–	10.9	-26.2	1.74	–	-6.4	-2.1	-0.076 [-0.17, +0.01]
	CA-HRP	1.276	–	11.3	-24.9	0.98	–	-5.3	-2.3	+0.004 [-0.09, +0.09]
	CA-HRP-IV	1.280	–	11.2	-26.6	1.91	–	-7.9	-2.1	–

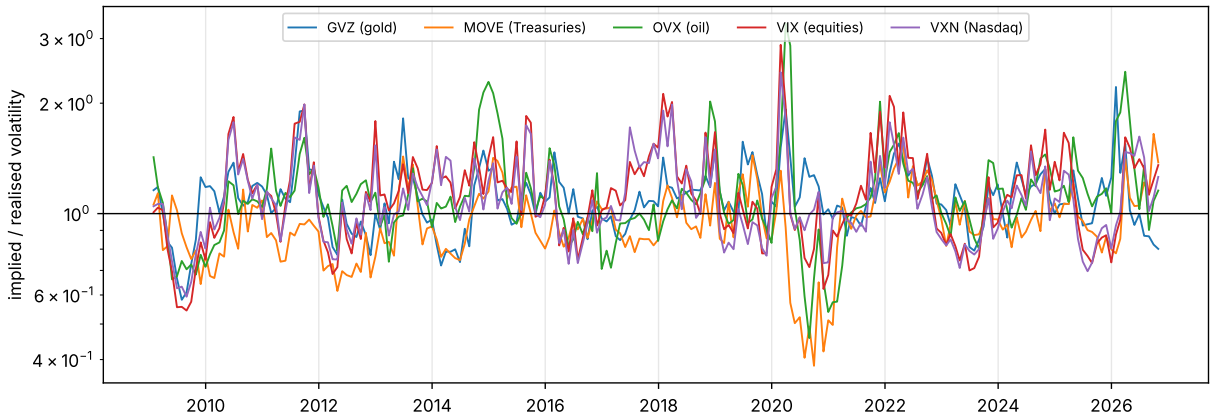


Figure 10: Implied-to-realised volatility ratios used by CA-HRP-IV in the multi-asset universe (log scale; 1 = implied equals realised).

8.5 Stricter statistics

Model Confidence Set. Table 16 applies the Model Confidence Set of Hansen et al. [6] (T_{max} statistic, 21-day block bootstrap, 1,000 draws, 90% level) to the six mandate-compliant methods. With negative returns as the loss, no method can be eliminated in any universe: realised returns cannot distinguish these methods over 15–25 years. With squared returns as the loss, constrained minimum variance is in the set in every universe, as it should be since it minimises variance by design; CA-HRP and CA-HRP-R are the only other methods that survive, and only in the multi-asset universe.

Deflated Sharpe ratio. During the project we evaluated 22 configurations of CA-HRP (design study, robust design study and ablations). Charging for all of them with the deflated Sharpe ratio of Bailey and López de Prado [1] (using the cross-method variance of Sharpe ratios, which is conservative), CA-HRP's Sharpe ratio remains significantly positive in every universe (DSR between 0.963 for countries and 1.000), despite strongly fat-tailed returns (daily kurtosis 9–20). Its positive risk-adjusted return is therefore not an artefact of trying many variants, although, as above, this is not evidence of superiority over the other compliant methods.

Risk calibration. Optimised portfolios tend to under-predict their own risk. We compare each method's predicted volatility at each rebalance with the realised volatility of the following month. Constrained minimum variance is the most optimistic in the stock universes (bias statistic 1.29 for NIFTY and 1.12 for US, against 1.21 and 1.05 for CA-HRP), consistent with the optimiser's curse; in the ETF universes the methods are similar (1.03–1.15). CA-HRP's risk forecasts are therefore at least as reliable as those of the optimisers it is compared with, which matters to a risk manager even when realised returns are indistinguishable.

8.6 Verification

Table 17 lists eleven automated checks run against the stored results. Our HRP implementation reproduces the independent PyPortfolioOpt implementation [13] to 10^{-16} on 36 windows. No look-ahead: replacing every return after a given date by random noise leaves all earlier weights and returns bit-for-bit unchanged. Re-running backtests reproduces the stored results exactly; transaction costs, Sharpe ratios, growth rates and drawdowns recomputed by independent code agree to 10^{-15}; the tables in the paper equal a fresh recomputation; three different solvers agree on the projection to 6×10^{-5}; and an audit of every stored portfolio of every compliant method finds no constraint breached by more than 2×10^{-8}. One check failed its strict threshold: our constrained

Table 16: Statistical checks. Left: membership of the 90% Model Confidence Set (✓) for return (negative return loss) and risk (squared return loss) among the mandate-compliant methods. Middle: CA-HRP's probabilistic Sharpe ratio against zero and its deflated Sharpe ratio after charging for 22 configurations tried. Right: risk calibration bias statistic (standard deviation of monthly returns standardised by predicted risk; 1 = calibrated, >1 = risk under-predicted).

	MCS (return)						MCS (risk)						CA-HRP		Bias statistic		
Universe	MinVar-C	MVO-C	HRP-Proj	CA-HRP-B	CA-HRP	CA-HRP-R	MinVar-C	MVO-C	HRP-Proj	CA-HRP-B	CA-HRP	CA-HRP-R	PSR	DSR	MinVar-C	HRP-Proj	CA-HRP
ETFs	✓	✓	✓	✓	✓	✓	✓				✓	✓	1.000	0.999	1.08	1.05	1.08
NIFTY	✓	✓	✓	✓	✓	✓	✓						1.000	1.000	1.29	1.21	1.20
US	✓	✓	✓	✓	✓	✓	✓						1.000	1.000	1.12	1.05	1.05
India ETFs	✓	✓	✓	✓	✓	✓	✓						1.000	1.000	1.13	1.14	1.13
US sectors	✓	✓	✓	✓	✓	✓	✓						0.999	0.998	1.00	0.97	0.98
Countries	✓	✓	✓	✓	✓	✓	✓						0.975	0.963	1.02	0.99	1.00

minimum-variance solutions have up to 1.8×10^{-4} relative excess variance over PyPortfolioOpt's, because daily variances are tiny relative to default solver tolerances. Re-solving with a rescaled objective changes that competitor's Sharpe ratio by at most 3×10^{-4}, so no conclusion depends on it; we keep the original numbers so that the repository reproduces every table exactly. A SHA-256 manifest of all data files is included in the repository.

Table 17: Verification suite (`scripts/verify.py`). Every check is run on the stored results; the one strict failure is reported together with its economic impact.

Check	What is measured	Value	Threshold	Result
Data manifest written	SHA-256 of every raw data file	0	0	pass
No look-ahead	weights up to a date unchanged when all later returns are replaced by noise	0	0	pass
Deterministic re-run	max diff vs stored results (2 universes × 3 methods)	0	1e-09	pass
Cost accounting	|(gross - net) - turnover × cost| summed per backtest, max	5.6e-16	1e-10	pass
Independent metric recomputation	max gap in Sharpe / CAGR / MaxDD, all backtests	4.4e-16	1e-09	pass
Paper tables match raw results	max gap Sharpe/Vol between table CSV and recomputation	2.2e-16	1e-12	pass
Raw constraint audit	largest breach of any constraint, 6 compliant methods × 6 universes, no tolerance	2.2e-08	1e-06	pass
Three solvers agree	max weight gap CLARABEL vs OSQP/SCS, 30 projections	5.7e-05	1e-04	pass
HRP matches PyPortfolioOpt	max |w_ours - w_ref| over 36 windows, 6 universes	1.1e-16	1e-08	pass
Min-variance matches PyPortfolioOpt	max relative excess variance of our solution over the reference, 36 windows	1.8e-04	1e-04	**fail**
Min-variance precision: economic impact	max change in MinVar-C Sharpe when re-solved with a rescaled objective, 6 universes	2.7e-04	1e-03	pass

9 Discussion

Five lessons emerge. First, *HRP is not deployable as-is* under realistic mandates: it breached the constraints at 48–100% of rebalances in all six universes, and the clipping fix used in practice still failed at up to 46% (27% on average with quarterly rebalancing). Second, *how* constraints are

enforced matters as much as *whether* they are: the same HRP target, projected with a risk-aware metric instead of a Euclidean one, gives a portfolio with 25% less volatility and a higher Sharpe ratio where constraints bind, lower or equal volatility in every universe, and the best true Sharpe ratio in a controlled simulation. The gain comes mostly from group constraints, where whole blocks of correlated weight must be replaced. Third, building constraints into the bisection, while elegant and provably feasible, is inferior to separating risk allocation from constraint enforcement. Fourth, neither the in-bisection approach nor a scenario-robust projection fixed the regime-change weakness; constrained minimum variance remains the method to beat on realised returns, and the Model Confidence Set cannot separate any compliant method on returns. Fifth, the case for CA-HRP rests on properties that hold broadly and can be proven: exact compliance, noise damping, interpretable compliance costs and well-calibrated risk. Its return advantage, by contrast, is specific to strongly binding, correlation-aligned group constraints, as the random-mandate and placebo study shows.

Limitations. The stock universes are survivorship-biased because they contain current large caps; this affects all methods equally. The gains of CA-HRP are concentrated in settings where constraints bind strongly; when they do not, it reduces to HRP, which is by design. Most Sharpe differences are not statistically significant at conventional levels, which is typical for portfolio comparisons over 15–20 years, and constrained minimum variance remains a strong benchmark. The tracking-error projection inherits the weakness of any covariance-based hedge, visible in 2022 when stock–bond correlation turned positive, and our regime-robust extension did not remove it. All data come from a single public source (Yahoo Finance), which needed one automated correction for unadjusted splits.

10 Conclusion

We showed that Hierarchical Risk Parity, despite its robustness, cannot be used directly under realistic portfolio mandates, and that the ad hoc fixes used in practice either fail to restore feasibility or distort the risk allocation. Constraint-Aware HRP separates the two problems: HRP decides how risk should be allocated, and a tracking-error projection finds the compliant portfolio closest to that allocation in risk terms. The method has clean theoretical properties, runs in milliseconds per rebalance, satisfied every constraint in more than 1,300 monthly rebalances across six markets (and at weekly and quarterly frequencies), improved risk-adjusted returns where constraints bind, including on a held-out test period, and was the best method in a controlled simulation. Evidence from three universes added after the design was frozen confirms that its gains scale with how hard the constraints bind. We also report two negative results that we believe are useful for the literature on hierarchical allocation: embedding constraints directly in the recursive bisection, although provably feasible, performs worse than projecting afterwards; and robustness to correlation regimes the estimation window has never seen cannot be obtained simply by adding covariance scenarios. Two further results explain the method's behaviour: the projection provably damps estimation noise while the common clipping fix can amplify it many times over, and the cost of compliance can be split exactly across the rules of a mandate, which tells a manager which rule is costing the most. A random-mandate study and a placebo test showed that the return advantage is specific to strongly binding, correlation-aligned group constraints, while the compliance and risk advantages hold broadly.

Future work includes projection metrics that learn regime changes in correlations, the use of CA-HRP with factor-model covariance estimates for universes of thousands of assets, and extending the capacity analysis to non-linear constraints such as tracking-error or CVaR budgets.

Code and data availability. All code, data and scripts that reproduce every table and figure are available at `https://github.com/pramukhjain123/capstone`.

References

[1] David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107, 2014.

[2] Heinz H. Bauschke and Patrick L. Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. Springer, 2nd edition, 2017.

[3] Vijay K. Chopra and William T. Ziemba. The effect of errors in means, variances, and covariances on optimal portfolio choice. *The Journal of Portfolio Management*, 19(2):6–11, 1993.

[4] Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953, 2009.

[5] Steven Diamond and Stephen Boyd. CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

[6] Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

[7] Ravi Jagannathan and Tongshu Ma. Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683, 2003.

[8] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[9] Olivier Ledoit and Michael Wolf. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008.

[10] Marcos López de Prado. Building diversified portfolios that outperform out of sample. *The Journal of Portfolio Management*, 42(4):59–69, 2016.

[11] Sébastien Maillard, Thierry Roncalli, and Jérôme Teïletche. The properties of equally weighted risk contribution portfolios. *The Journal of Portfolio Management*, 36(4):60–70, 2010.

[12] Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[13] Robert Andrew Martin. PyPortfolioOpt: Portfolio optimization in Python. *Journal of Open Source Software*, 6(61):3066, 2021.

[14] Richard O. Michaud. The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal*, 45(1):31–42, 1989.

[15] Johann Pfitzinger and Nico Katzke. A constrained hierarchical risk parity algorithm with cluster-based capital allocation. Stellenbosch Working Paper Series WP14/2019, 2019.

[16] Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

[17] Thomas Raffinot. Hierarchical clustering-based asset allocation. *The Journal of Portfolio Management*, 44(2):89–99, 2017.

[18] Thomas Raffinot. The hierarchical equal risk contribution portfolio. SSRN Working Paper 3237540, 2018.

[19] Richard Roll. A mean/variance analysis of tracking error. *The Journal of Portfolio Management*, 18(4):13–22, 1992.

[20] Lloyd S. Shapley. A value for n-person games. In H. W. Kuhn and A. W. Tucker, editors, *Contributions to the Theory of Games II*, pages 307–317. Princeton University Press, 1953.

[21] Florin Spinu. An algorithm for computing risk parity weights. SSRN Working Paper 2297383, 2013.

A Design study on the validation period

Table 18 lists the eight configurations compared before the method was frozen, using only data up to the end of the validation window. The tracking-error projection was best in the multi-asset universe and tied for best in the stock universes. The sample-covariance version scored marginally higher than the Ledoit–Wolf one; we kept Ledoit–Wolf, chosen beforehand for its better conditioning, since the difference is within noise. The initial design ("cap+tree (v0)") was the weakest in the multi-asset universe.

Table 18: Design study on the validation period only (net of 10 bps). "cap" = capacity-aware bisection; "tree" = tree-weighted projection; "TE" = tracking-error projection. LW covariance unless stated.

	Validation Sharpe			ETFs	
Variant	ETFs	NIFTY	US	Vol	Turn.
cap+tree (v0)	0.605	1.737	0.639	10.2	2.59
euclid (LW)	0.675	1.731	0.645	10.2	1.07
cap+euclid	0.637	1.737	0.646	10.3	2.11
diag-var proj	0.743	1.731	0.648	9.5	1.13
TE proj	0.878	1.738	0.651	7.3	1.19
cap+TE proj	0.750	1.737	0.651	8.6	2.45
TE+tree proj	0.749	1.741	0.651	7.9	2.30
TE proj (sample cov)	0.887	1.745	0.655	7.3	1.11

Table 19 shows the corresponding study for the regime-robust variant, run later on the same validation windows. A first run of this study contained a bug: the hedges-fail stress matrix was not positive semi-definite, so the solver failed and returned the infeasible HRP target. We repaired the matrix (eigenvalue clipping), added a feasible fallback and a unit test, and reran the study; the selection was unchanged.

Table 19: Validation-period selection of the CA-HRP-R scenario set (Sharpe, net of 10 bps). The robust variant with the best mean was adopted before any test-period or external-universe result was computed.

Variant	ETFs	NIFTY	US	Mean
CA-HRP (reference)	0.878	1.738	0.651	1.089
max{long,short,stress}	0.806	1.736	0.649	1.063
max{long,short}	0.865	1.736	0.649	1.083
max{long,stress}	0.768	1.738	0.651	1.052
mean{long,short,stress}	0.779	1.737	0.650	1.055

B Turn-of-month returns in the NIFTY universe

Daily returns on the last trading day of the month are unusually large in the NIFTY sample (an average of 0.42% for an equal-weight portfolio up to 2018, against 0.07% on other days). All methods trade at the month-end close and earn that day's return with their pre-trade weights, so the effect raises all NIFTY Sharpe ratios similarly and does not change their ranking.